

HF Pulse Signal Injection Based Method for Sensorless Control of PMSM with Cogging Torque Compensation

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Abstract—High-frequency (HF) signal injection methods are one of the key techniques for accurately estimating the rotor position of permanent magnet synchronous machines (PMSMs). The rotor position is determined by superimposing HF signals onto the fundamental control signals, producing a combined signal that serves as a reference for the PWM modulator. The rotor position is then extracted from the current response caused by the injected signal. This paper introduces a modification to the HF pulse signal injection method for sensorless control. The proposed approach improves control dynamics by reducing the signal injection period from two control periods to just one, and by adding the cogging torque compensation, thereby minimizing current and speed ripple. Experimental results obtained with a 2.5 kW PMSM and RT-LAB software obtained for the 1% of nominal speed validate the effectiveness of the proposed method.

Index Terms—PMSM, sensorless control, signal injection, cogging torque

I. INTRODUCTION

Position sensorless control techniques for permanent magnet synchronous machines (PMSMs) have been recently successfully commercialized for several application areas such as ventilation, pumping, home appliances or wind power generators. Elimination of the rotor position sensor from the PMSM control structure will always remain desirable as it offers replacement of hardware, which is a potential source of failures, with a pure software-based sensorless solution.

Position control methods of PMSM suitable for low-speed machine operation can be categorized into two big groups: fundamental pulse-width modulation excitation-based (FPE) methods and high-frequency (HF) signal injection-based methods [1]. Some well-known examples from the first group are the INFORM method [2] and the zero-voltage vector (ZVV) method [3]. The common feature of ZVV methods is that the

normal PWM switching pattern is modified by adding appropriately timed additional zero voltage vector. By measuring and analyzing the rate of current decay during these zero-voltage periods, the rotor position can be estimated. These methods offer several advantages, including straightforward algorithm, low acoustic noise, insensitivity to parameter variations and improved rotor speed/position estimation accuracy. However, they require additional or specialized hardware such as balanced resistor network, extra-high resolution current sensors or special current differential detectors [4], [5]. On the other side, the HF signal injection-based methods operate on the principle that rotor position information can be effectively extracted by injecting of HF signal and by measuring of current responses that will arise as a response to the signal injection. The rotor position is obtained from the current response caused by the injected voltage signal.

The HF signal injection-based methods can be roughly divided into following categories [6]:

- HF rotating sinusoidal signal injection-based method [7], in which sinusoidal voltages are injected to stationary reference frame,
- HF pulsating sinusoidal signal injection-based method, in which sinusoidal voltages are injected to dq reference frame voltages [8], [9], [10],
- HF square-wave signal injection-based method (HFSWI), in which square-wave voltage is injected in dq reference frame voltages [11], [12], [13],
- and HF pulse signal injection-based method (HFPSI), in which square-wave voltage is injected in dq reference frame while the field-oriented control is interrupted [14], [15].

The advantage of the HFSWI and HFPSI methods over sinusoidal signal-injection based methods lies in the fact the

frequency of the injected square-wave signal can be much higher than the frequency of the injected sinusoidal signal, what offers various benefits. The inherent time delay caused by using digital filtering needed for older methods is minimized by applying the HFSWI or HFPSI methods. It results in higher bandwidth of both position observer and PI speed controller.

When comparing HFSWI and HFPSI methods, the main difference lies in following. In the HFSWI, the injected HF square-wave voltage is superimposed to dq reference voltage given by the output of the PI controller. However, the HFPSI is based on separating the HF voltage injection signal from the field-oriented control (FOC) algorithm. Both the HF signal injection and the FOC algorithm operate within their own dedicated control periods.

The original contribution of this study is as follows: for implementation of HFPSI method [14], two switching periods are necessary to eliminate the negative influence of the inverter nonlinearities on the rotor position estimation. During these two switching periods the motor torque is not controlled and negative effects may emerge under full load conditions. The solution for this problem is proposed in this paper.

II. BRIEF OVERVIEW OF HF SIGNAL INJECTION THEORY FOR HFSWI AND HFPSI

A. HF Square Wave Injection Method

The mathematical model of interior permanent magnet synchronous machine (IPMSM) in the stationary reference frame (SRF) $\alpha - \beta$ is given as:

$$\begin{bmatrix} u_\alpha \\ u_\beta \end{bmatrix} = \begin{bmatrix} R_s & 0 \\ 0 & R_s \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + \frac{d}{dt} \begin{bmatrix} \Psi_\alpha \\ \Psi_\beta \end{bmatrix}. \quad (1)$$

where R_s is the stator resistance, u_α, u_β - supply voltage vector components, i_α, i_β - stator current vector components, Ψ_α, Ψ_β - stator flux vector components.

Stator flux vector components can be calculated using the following formula:

$$\begin{bmatrix} \Psi_\alpha \\ \Psi_\beta \end{bmatrix} = \begin{bmatrix} L_\Sigma + L_\Delta \cos(2\theta_e) & L_\Delta \sin(2\theta_e) \\ L_\Delta \sin(2\theta_e) & L_\Sigma - L_\Delta \cos(2\theta_e) \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + \begin{bmatrix} \cos(\theta_e) & -\sin(\theta_e) \\ \sin(\theta_e) & \cos(\theta_e) \end{bmatrix} \begin{bmatrix} \Psi_{pm} \\ 0 \end{bmatrix}, \quad (2)$$

where Ψ_{pm} is the permanent magnet flux linkage, θ_e - rotor position, $L_\Sigma = \frac{L_d + L_q}{2}$, $L_\Delta = \frac{L_d - L_q}{2}$, and L_d and L_q are d and q -axis inductances; respectively.

If the HF voltage signal is injected the model given by (1)-(2) can be simplified. When the frequency of the injected voltage signal is high enough, a voltage drop on R_s can be neglected, and at low speed operation all terms containing speed ω_e can be approximated by zero. After some manipulation, HF current derivatives, denoted below as subscripted h , can be found:

$$\frac{d}{dt} \begin{bmatrix} i_{\alpha h} \\ i_{\beta h} \end{bmatrix} = c \begin{bmatrix} L_\Sigma - L_\Delta \cos(2\theta_e) & -L_\Delta \sin(2\theta_e) \\ -L_\Delta \sin(2\theta_e) & L_\Sigma + L_\Delta \cos(2\theta_e) \end{bmatrix} \begin{bmatrix} u_{\alpha h} \\ u_{\beta h} \end{bmatrix}, \quad (3)$$

where $c = \frac{1}{L_\Sigma^2 - L_\Delta^2}$.

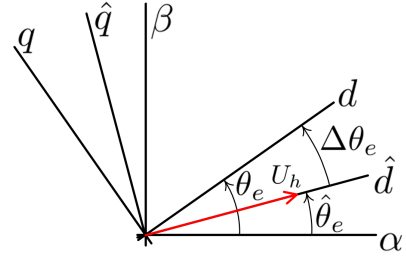


Fig. 1. Synchronous reference frame and estimated $\hat{d}\hat{q}$ reference frame.

The HF voltage is injected into the estimated rotor reference frame (ERRF) as shown in Fig. 1, where dq -axes are the actual rotor reference frame (RRF) and $\hat{d}\hat{q}$ -axes are estimated RRF. The position estimation error is defined as $\hat{\theta}_e = \theta_e - \hat{\theta}_e$. The current derivatives of the HF model in the estimated RRF can be obtained via coordinate transformation [6] resulting in:

$$\frac{d}{dt} \begin{bmatrix} i_{\hat{d}h} \\ i_{\hat{q}h} \end{bmatrix} = c \begin{bmatrix} L_\Sigma - L_\Delta \cos(2\hat{\theta}_e) & -L_\Delta \sin(2\hat{\theta}_e) \\ -L_\Delta \sin(2\hat{\theta}_e) & L_\Sigma + L_\Delta \cos(2\hat{\theta}_e) \end{bmatrix} \begin{bmatrix} u_{\hat{d}h} \\ u_{\hat{q}h} \end{bmatrix}, \quad (4)$$

where $u_{\hat{d}h}, u_{\hat{q}h}, i_{\hat{d}h}, i_{\hat{q}h}$ are HF voltages and currents in the ERRF.

In HFSWI the voltage is injected to estimated $\hat{d}$ -axis and the injection voltage is defined as:

$$\begin{bmatrix} u_{\hat{d}h} \\ u_{\hat{q}h} \end{bmatrix} = \begin{bmatrix} (-1)^k U_h \\ 0 \end{bmatrix}, \quad (5)$$

where U_h is amplitude and k is the control step, respectively. By substituting (5) into (4), the HF current variation in a sampling period ΔT can be found:

$$\begin{bmatrix} \Delta i_{\hat{d}h} \\ \Delta i_{\hat{q}h} \end{bmatrix} = c u_{\hat{d}h} \Delta T \begin{bmatrix} L_\Sigma - L_\Delta \cos(2\hat{\theta}_e) \\ -L_\Delta \sin(2\hat{\theta}_e) \end{bmatrix}. \quad (6)$$

If the injection frequency is high, the fundamental current component can be considered constant during sampling period, thus a variation of current measured in q -axis approximates to that of HF current $\Delta i_{\hat{q}h} \approx i_{\hat{q}}(k) - i_{\hat{q}}(k-1)$. For small values of the position estimation error, a signal ϵ associated with the position estimation error, can be found:

$$\epsilon \approx \frac{\Delta i_{\hat{q}}}{u_{\hat{d}h}} = -2cL_\Delta \Delta T \hat{\theta}_e. \quad (7)$$

The ϵ acts as an input to the saliency-tracking observer and by driving ϵ to zero, the saliency-tracking observer forces the convergence of the estimated position to the actual one. It can be seen from (7) that the estimated position error $\hat{\theta}_e$ is minimized by changes in q -axis current.

B. HF Pulse Signal Injection Method

By separating the FOC period from the injection period, the fundamental current component does not interfere with the HF component induced by voltage injection. The current values sampled during the HF injection period in the HFPSI method consist exclusively of HF current responses, which is a significant advantage. This technique of separating injection

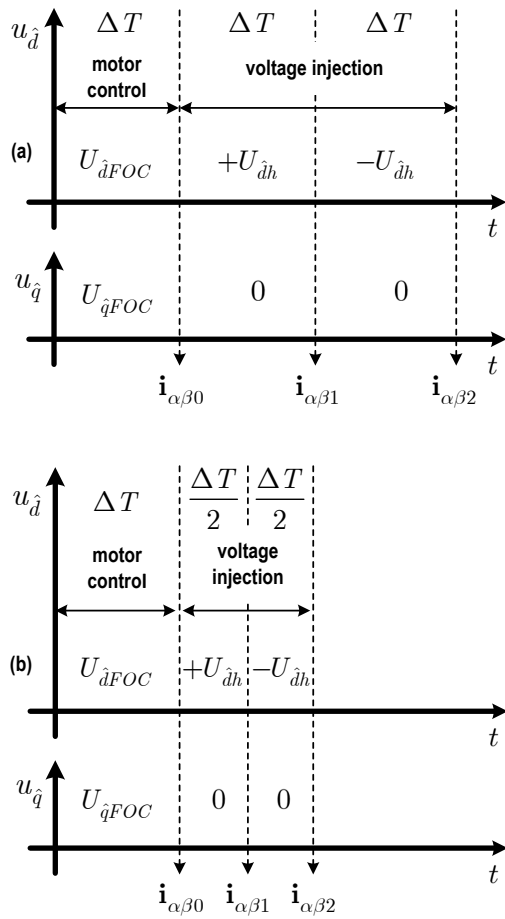


Fig. 2. a) Voltage injection periods of the original HFPSI method [15], where the motor is controlled once every three switching periods. b) Voltage injection periods of the proposed modification, enabling motor control every other switching period.

and control period is known as high-frequency pulse signal injection (HFPSI). The sampled currents in injection period are HF current responses only. This method was first introduced in [14] where the motor control period was followed by a single voltage injection period. In [1] the negative influences of the voltage errors caused by inverter nonlinearities were discussed. These voltage errors can be effectively suppressed by injection positive and negative voltage pulse.

During the injection period, two consecutive voltage vectors are applied to the motor windings. In the first half the positive voltage pulse $\mathbf{u}_{\hat{d}h-p}$ and in the second half the negative voltage pulse $\mathbf{u}_{\hat{d}h-n}$ is injected into the observed $\hat{d}$ -axis (see Fig. 2a), while the voltage in the observed $\hat{q}$ -axis is kept zero:

$$\begin{bmatrix} u_{\hat{d}h-p} \\ u_{\hat{q}h-p} \\ u_{\hat{d}h-n} \\ u_{\hat{q}h-n} \end{bmatrix} = \begin{bmatrix} U_h \\ 0 \\ -U_h \\ 0 \end{bmatrix}, \quad (8)$$

where $\mathbf{u}_{\hat{d}h-p}$, $\mathbf{u}_{\hat{d}h-n}$ and $\mathbf{u}_{\hat{q}h-p}$, $\mathbf{u}_{\hat{q}h-n}$ are the HF voltages injected to the observed $\hat{d}$ -axis and $\hat{q}$ -axis during the positive and negative injection sequence, respectively. The inverter

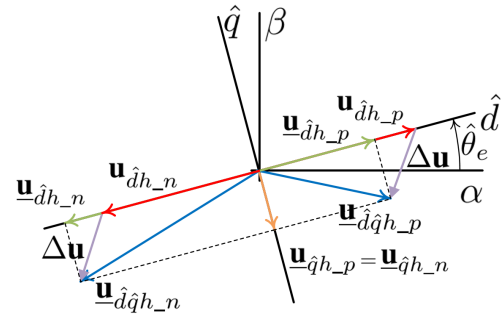


Fig. 3. Coordinate frames and distorted voltages for positive and negative injection periods.

nonlinearities introduced by non-ideal switches, dead-times, voltage drops, etc., may cause distortions leading to an offset of the injected voltage vector. Therefore, assuming the voltage errors will not change during two consecutive switching periods, real voltages applied to the motor windings can be expressed as:

$$\begin{bmatrix} \mathbf{u}_{\hat{d}h-p} \\ \mathbf{u}_{\hat{q}h-p} \\ \mathbf{u}_{\hat{d}h-n} \\ \mathbf{u}_{\hat{q}h-n} \end{bmatrix} = \begin{bmatrix} U_h - \Delta u_{\hat{d}h-p} \\ -\Delta u_{\hat{q}h-p} \\ -U_h - \Delta u_{\hat{d}h-p} \\ -\Delta u_{\hat{q}h-p} \end{bmatrix}, \quad (9)$$

where $\mathbf{u}_{\hat{d}h-p}$, $\mathbf{u}_{\hat{q}h-p}$, $\mathbf{u}_{\hat{d}h-n}$, $\mathbf{u}_{\hat{q}h-n}$ are the HF voltages including the inverter voltage errors. The resulting HF voltage vector can be formed into a more compact form as follows:

$$\begin{bmatrix} \mathbf{u}_{\hat{d}h-p} \\ \mathbf{u}_{\hat{d}h-n} \end{bmatrix} = \begin{bmatrix} U_h - \Delta \mathbf{u} \\ -U_h - \Delta \mathbf{u} \end{bmatrix}, \quad (10)$$

where $\mathbf{u}_{\hat{d}h-p}$ and $\mathbf{u}_{\hat{d}h-n}$ are the final HF voltage vectors during the positive and negative injection sequences and $\Delta \mathbf{u}$ is the voltage error in estimated rotor reference frame. The coordinate distribution for the injected HF voltage vectors assuming voltage errors is shown in Fig. 3. The inverter voltage error between two consecutive injection periods is assumed to be the same. The influence of the voltage error can be canceled by adding two current differences measured during the positive and negative HF voltage injection.

For the current deviations in the stator reference frame stands:

$$\begin{bmatrix} \Delta i_{\alpha} \\ \Delta i_{\beta} \end{bmatrix} = \begin{bmatrix} \Delta i_{\alpha-p} - \Delta i_{\alpha-n} \\ \Delta i_{\beta-p} - \Delta i_{\beta-n} \end{bmatrix} = \begin{bmatrix} -i_{\alpha 0} + 2i_{\alpha 1} - i_{\alpha 2} \\ -i_{\beta 0} + 2i_{\beta 1} - i_{\beta 2} \end{bmatrix}, \quad (11)$$

where $i_{\alpha\beta 0}$, $i_{\alpha\beta 1}$ and $i_{\alpha\beta 2}$ are the currents sampled at the beginning, middle, and end of the voltage injection period (as shown in Fig. 2b); while $\Delta i_{\alpha-p}$, $\Delta i_{\beta-p}$ while $\Delta i_{\alpha-n}$, $\Delta i_{\beta-n}$ are the current differences in the stator reference frame for the positive and negative injection sequences, respectively. They can be transformed into the observed rotor reference frame by Park transformations:

$$T_{\Delta i_{\alpha\beta} \rightarrow \Delta i_{\hat{d}\hat{q}}} = \begin{bmatrix} \cos(\hat{\theta}_e) & \sin(\hat{\theta}_e) \\ -\sin(\hat{\theta}_e) & \cos(\hat{\theta}_e) \end{bmatrix}, \quad (12)$$

Therefore, the current derivations related to the observed position error $\tilde{\theta}_e$ can be expressed as follows:

$$\begin{bmatrix} \Re(\Delta \mathbf{i}_{\hat{d}\hat{q}}) \\ \Im(\Delta \mathbf{i}_{\hat{d}\hat{q}}) \end{bmatrix} = \begin{bmatrix} \Delta i_\alpha \cos(\hat{\theta}_e) + \Delta i_\beta \sin(\hat{\theta}_e) \\ -\Delta i_\alpha \sin(\hat{\theta}_e) + \Delta i_\beta \cos(\hat{\theta}_e) \end{bmatrix}. \quad (13)$$

For small values of the position estimation error, the signal ϵ associated with the position estimation error is given by the sum of imaginary parts of the current variations from positive and negative voltage injection:

$$\epsilon = \Im(\Delta \mathbf{i}_{\hat{d}\hat{q}}) = \sin(\tilde{\theta}_e) \approx \tilde{\theta}_e, \quad (14)$$

and then enters the saliency tracking observer.

III. PROPOSED MODIFICATION OF HFPSI METHOD

The HFPSI method is based on separating the HF voltage injection from the motor control. Both of them have a dedicated period of the same length. During the FOC period the HF injection is stopped, and during the HF injection period the motor control is interrupted. In original HFPSI method two opposite voltage vectors are injected to reduce the effect of voltage errors on position estimation. Thus, one motor control period is followed by two voltage injection periods where a positive and negative HF pulses are injected. The HF injection and motor control periods for original HFPSI method are shown in Fig. 2a. The motor torque is controlled in every third switching period what may cause pulsating of torque.

A modified approach is introduced here with the use of double update PWM method to improve dynamic performance and reduce torque fluctuation. The motor control is interrupted only for a single switching period while the compensation of inverter nonlinearities by injecting two opposite voltage components is still used, as shown in Fig. 2b.

The currents $i_{\alpha\beta 0}$, $i_{\alpha\beta 1}$ and $i_{\alpha\beta 2}$ are sampled at the beginning, in the middle, and at the end of the voltage injection period. When motor control is interrupted, these currents serve as direct high-frequency responses and will be used to calculate current differences. The sample $i_{\alpha\beta 2}$ will also serve as a feedback in motor control for the current controllers.

The current responses captured on oscillogram are presented in Fig. 4. For this demonstration, only zero vectors were applied during the FOC period, as evident from the switching patterns. Therefore, no change occurs in the phase current. In Fig. 4a, the two subsequent injection periods are equal in length to the FOC period. First, a positive voltage is applied, causing the phase current to change by approximately 1000 mA. This is followed by a negative voltage pulse, which restores the phase current to its pre-injection value, allowing a new FOC period to be executed. The modification is presented in Fig. 4b. One injection period is only half the length of the FOC period. During the positive voltage injection the current change is around 500 mA (note the current probe resolution was changed) since the duration of the injection is shorter.

The original method of HFPSI does not employ digital filters, resulting in increased regulator bandwidth and improved dynamic performance compared to other HF injection techniques. However, uncontrolled motor torque during the

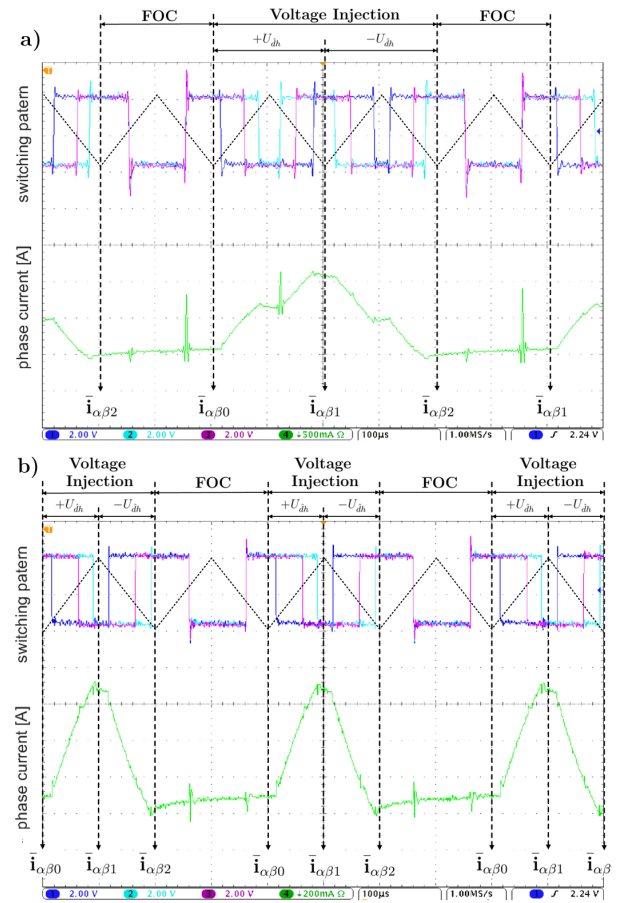


Fig. 4. Transients of the upper transistor switching signals and phase current under the HF injection: a) conventional approach, b) proposed approach.

voltage injection sequence introduces unwanted torque ripples and extends the overall FOC response time. These ripples can be even higher when using a machine that has a relatively high cogging torque. The generated torque ripple negatively affect the motor performance and can contribute to motor vibration in light load and low speed applications.

In this article, a combination of modified HFPSI method with a cogging torque compensation based on q -axis current and rotor position mapping [16] is presented. Firstly, a mapping of the q -axis current to the rotor position must be established. This is achieved by using a position sensor connected to the rotor shaft while it completes one full mechanical revolution. When the machine operates at a very slow, constant speed without load, the measured q -axis current can be shown to correlate with the motor's cogging torque. By storing this relationship in a lookup table (LUT), a simple cogging torque compensator is established. In full sensorless control mode, the observed rotor position is used as input to the LUT, with the output providing the q -axis current component required for cogging torque compensation, as shown in Fig. 5. It is important to note that the effectiveness of cogging torque compensation depends on the type of sensorless algorithm employed.

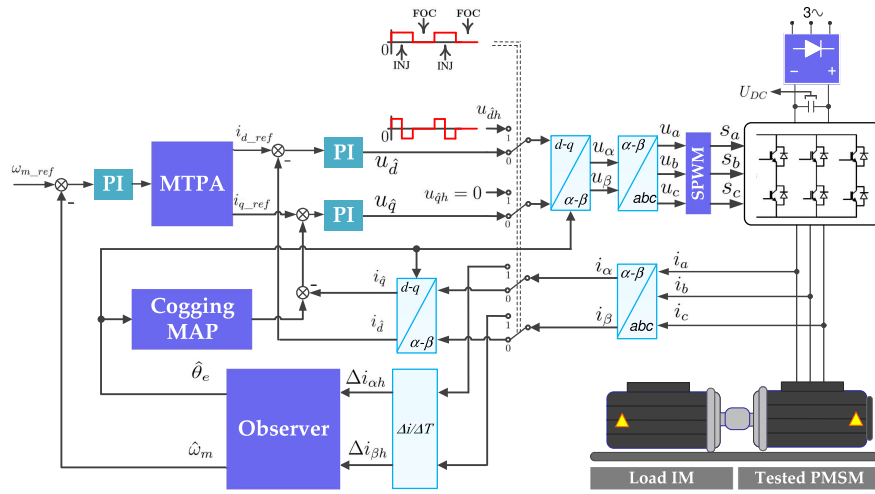


Fig. 5. Block diagram of sensorless FOC based on HF pulse voltage injection and cogging torque compensation.

IV. EXPERIMENTAL RESULTS

The control algorithm was developed in the MATLAB/Simulink environment using RT-LAB software. The OP 5600 hardware-in-the-loop (HIL) simulator served as the real-time controller. Its digital outputs directly drove the gate signals of the voltage source inverter in the INFINEON EVAL-M1-IM818-A development kit. The control period for was set to $\Delta T = 120\mu s$ with sampling time $T_s = 10\mu s$. Currents were measured using CKSR LEM transducers and processed by a 16-bit A/D converter. The PMSM machine with parameters in Table I was loaded by an induction machine controlled by industrial converter. The block diagram of the motor control, along with the HFPSI-based position observer, is shown in Fig. 5. The $\alpha\beta$ currents were measured in the time instants as shown in Fig. 4b and used as inputs to the saliency tracking observer shown in Fig. 6.

Experimental results for full sensorless control, where observed position and speed were used as feedback values, are shown in Fig. 7 and Fig. 8. The effect of cogging torque compensation can be observed from Fig. 7 where considerable speed ripple reduction was achieved. The comparison of

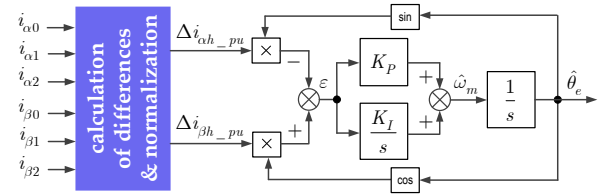


Fig. 6. Saliency-tracking observer for Fig. 5.

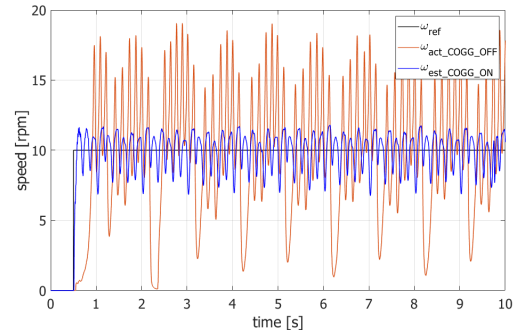


Fig. 7. The effect of cogging torque compensation on sensorless speed control.

TABLE I
IPMSM PARAMETERS AND CONTROLLER PARAMETERS

rated power	$P_n = 2.53 \text{ kW}$
rated torque	$M_n = 10.5 \text{ Nm}$
rated voltage	$U_n = 380 \text{ V}$
rated current	$I_n = 4.2 \text{ A}$
torque constant	$k_t = 2.5 \text{ Nm/A}$
no. of pole pairs	$2p = 6$
rated speed	$n_n = 2300 \text{ rpm}$
stator resistance	$R_{ab} = 2.92 \text{ } \Omega$
d -axis inductance	$L_d = 0.011 \text{ H}$
q -axis inductance	$L_q = 0.031 \text{ H}$
injected voltage	$U_h = 40 \text{ V}$
PI speed controller	$K_{P\omega} = 0.1, K_{I\omega} = 0.5$
PI current controllers	$K_{Pi} = 30, K_{Ii} = 500$
PLL parameters	$K_P = 200, K_I = 10 \text{ 000}$

original HFPSI [14] and modified HFPSI method are shown in Fig. 8. For both experiments, the reference speed was set to the 1% of nominal speed and the step load torque was applied. The modified HFPSI method offers comparable speed and position estimation precision while reducing current ripple. This is achieved by controlling motor torque every other step ensuring more frequent adjustments unlike the original method, which controls torque every third step.

V. CONCLUSION

The original HFPSI method employs two consecutive control periods for injection of positive and negative voltage pulse to eliminate inverter nonlinearities. In the proposed method, by

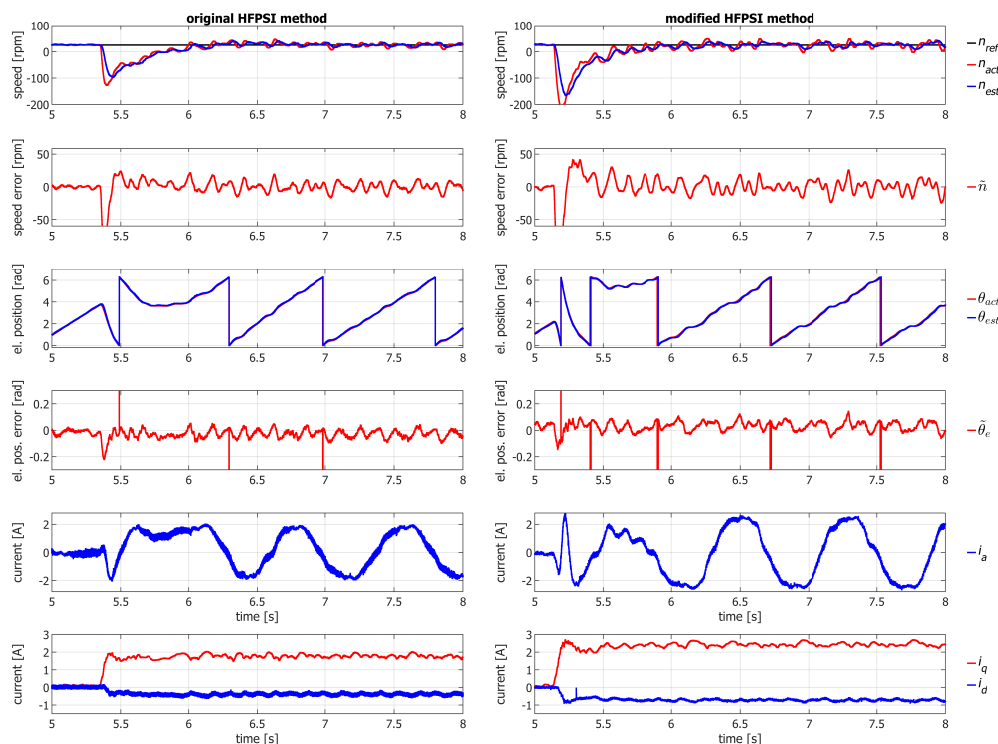


Fig. 8. Experimental results - comparison of original and modified HFPSI method under loading. The speed reference is 25 rpm (approx. 1% of rated speed.)

employing double update PWM mode, only one control period for injection is needed. Experimental validation on a 2.5 kW PMSM demonstrated improvement of proposed method over the original one for current ripple reduction. Future research should focus on analyzing cogging torque compensation using various sensorless approaches.

VI. ACKNOWLEDGMENT

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